

TARGET-90

JKBOSE CLASS-11TH

SETS

One Shot

MATHS

Mehraj sir



Topics To Be Covered

- 1 ... Sets (Definition)
- 2 ... Important Question.
- 3 ...

Types of Sets

- 1 Empty Set (Null Set)**
A set with no elements.
Example: $A = \{ \}$ or $A = \emptyset$
- 2 Finite Set**
A set with a finite number of elements.
Example: $B = \{1, 2, 3, 4\}$
- 3 Infinite Set**
A set with infinite number of elements.
Example: $N = \{1, 2, 3, \dots\}$
- 4 Equal Sets**
Two sets are equal if they have the same elements.
Example: $A = \{1, 2, 3\}$ and $B = \{1, 2, 3\}$

SETS

A set is a well-defined collection of distinct objects.

$$A = \{1, 2, 3, 4\}$$

$$B = \{2, 4, 6, 8\}$$

$$C = \{1, 3, 5, 7\}$$

Elements of a set
are written within
curly brackets $\{ \}$.

Subsets

- A set A is a subset of set B ($A \subseteq B$) if every element of A is an element of B.

Example:

$A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$
Then, $A \subseteq B$

Proper Subset

$A \subset B$ if $A \subseteq B$ and $A \neq B$.

Example: $A = \{1, 2\}$ and $B = \{1, 2, 3\}$
Then, $A \subset B$

Power Set

- The power set of a set A is the set of all subsets of A.

Example:

If $A = \{1, 2\}$, then

$$P(A) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}$$

Number of elements in power set = 2^n

(where n is the number of elements in the set)

Union of Sets

- $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Example:

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$



$A \cup B$

Intersection of Sets

- $A \cap B = \{x : x \in A \text{ and } x \in B\}$

Example:

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

$A \cap B = \{3\}$



$A \cap B$

Difference of Sets

- $A - B = \{x : x \in A \text{ and } x \notin B\}$

Example:

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

$A - B = \{1, 2\}$



$A - B$

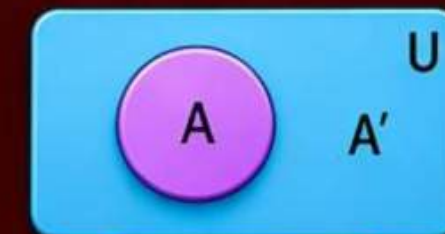
Complement of a Set

- $A' = \{x : x \in U \text{ and } x \notin A\}$
(where U is the universal set)

Example:

If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2\}$

Then, $A' = \{3, 4, 5\}$



Complement

Venn Diagrams



Set A



Disjoint Sets
($A \cap B = \emptyset$)



Intersecting Sets
($A \cap B \neq \emptyset$)



Subset
($A \subset B$)

Sets form the
foundation of modern
mathematics!

INTRODUCTION

★ **Definition:** A well defined collection of objects is known as a Set.

* Well defined $\rightarrow$ Proper definition with the help of which we can determine whether an object belongs to the collection or not.

(Same Meaning for everyone)

- * Beautiful
- * Good
- * Bad
- * Intelligent
- * Ugly

(Not well defined)

* Sets are denoted by Capital letters.

* Repetition of objects is not allowed in a Set.

* $\mathbb{N} \rightarrow$ Set of Natural Numbers

* $\mathbb{W} \rightarrow$ Set of Whole Numbers

* $\mathbb{Z} \rightarrow$ Set of Integers

* $\mathbb{Q} \rightarrow$ Set of Rational numbers

* $\mathbb{R} \rightarrow$ Set of Real numbers

* $\mathbb{C} \rightarrow$ Set of Complex numbers

* $\mathbb{Z}^+ \rightarrow$ Set of Positive Integers

* $\mathbb{R}^+ \rightarrow$ Set of Positive real numbers.

* The members of the Set are known as the elements of the Set.

* The number of elements in a Set is known as the Cardinality of the Set.

QUESTION



A Set is a

- 1 Collection of Objects ~~X~~
- 2 Collection of Numbers ~~X~~
- 3 Well defined Collection of Objects ✓✓
- 4 Well defined Collection of Numbers ~~X~~

QUESTION



The set of Rational numbers is Usually denoted by

1 Q ✓

2 R ✗

3 Z ✗

4 N ✗

QUESTION



Which of the following Collections is not a Set?

- 1 The collection of Natural numbers between 1 and 20 ✕
- 2 The collection of Roots of the equation $x^2 - 5x + 6 = 0$ ✕
3, 2
- 3 The collection of Prime numbers between 1 and 20 ✕
- 4 The collection of Intelligent Students of J & K ✓✓

QUESTION



If A denotes the **set of Prime numbers between 1 and 30**, then the **cardinality of A** is equal to

$|A|$ or $n(A)$ ✓

2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

1

8 ✗

2

10 ✓

3

9 ✗

4

11 ✗

$$n(A) = 10.$$

QUESTION



If B denotes the **set of Vowels in English Alphabets** then the **cardinality of B** is equal to

a, e, i, o, u

$$n(B) = 5$$

1

5 ✓

2

4 ✗

3

21 ✗

4

8 ✗

Representation of a Set

Roaster form
/ Tabular form

* In this form, we list all the elements of the set, separate them by comma's & are enclosed within Bracket.

$$A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$$

Set Builder form
/ Property Method

* In this form, we represent all the elements of the set by means of a variable (variables) followed by the property satisfied by each element of the set.

$$A = \{x : x \text{ is a prime number b/w } 1 \text{ \& } 30\}$$

EXAMPLE



Write down the following sets in the set-Builder form:

- (a) $\{3, 6, 9, 12\} \longrightarrow \{x : x = 3n, n \in \mathbb{N} \text{ and } 1 \leq n \leq 4\}$
- (b) $\{2, 4, 8, 16, 32\} \longrightarrow \{x : x = 2^n, n \in \mathbb{N} \text{ and } 1 \leq n \leq 5\}$
 $\begin{matrix} 2^1 & 2^2 & 2^3 & 2^4 & 2^5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 4 & 8 & 16 & 32 \end{matrix}$
- (c) $\{1, 4, 9, 16, \dots, 100\} \longrightarrow \{x : x = n^2, n \in \mathbb{N} \text{ and } 1 \leq n \leq 10\}$
 $\begin{matrix} 1^2 & 2^2 & 3^2 & 4^2 & \dots & 10^2 \\ \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow \\ 1 & 4 & 9 & 16 & & 100 \end{matrix}$

QUESTION



The Set of all Letters of the Word **"MATHEMATICS"** in roster form is :

$\{M, A, T, H, E, I, C, S\}$

1 ~~$\{M, T, A, H, M, I, C, S\}$~~

2 ~~$\{M, A, T, H, E, T, I, C, S\}$~~

3 $\{A, M, T, E, H, I, C, S\}$

4 ~~$\{MA, TH, E, I, C, S\}$~~

QUESTION



The Set $A = \{x : x \text{ is Positive prime} < 10\}$ in the tabular form is

- 1 $\{1, 2, 3, 5, 7\}$ ✗
- 2 $\{1, 3, 5, 7, 9\}$ ✗
- 3 $\{2, 3, 5, 7\}$ ✓✓
- 4 None of these ✗

QUESTION



Which of the following is a set Builder form of the set $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right\}$?

1 $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } n > 1\right\}.$

2 $\left\{x : x = \frac{n-1}{n+1}, n \in \mathbb{N} \text{ and } n > 2\right\}.$ *H/W*

3 $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 < n < 2\right\}.$

4 $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 \leq n < 4\right\}.$

QUESTION



If $A = \{x : x^2 - x - 6 = 0 \text{ and } x \in \mathbb{N}\}$, then the set A in Roster form is

1 $\{3\}$

h/w

2 $\{2\}$

3 $\{3, -2\}$

4 $\{1, 2\}$

QUESTION



List all the elements of the following sets:

(i) $A = \{x : x \text{ is an odd natural number}\} = \{1, 3, 5, \dots\}$

(ii) $B = \{x : x \text{ is an integer, } -\frac{1}{2} < x < \frac{9}{2}\} = \{0, 1, 2, 3, 4\}$

(iii) $C = \{x : x \text{ is an integer, } x^2 \leq 2\} = \{-1, 0, 1\}$

(iv) $D = \{x : x \text{ is a letter in the word "LOYAL"}\} = \{L, O, Y, A\}$

(v) $E = \{x : x \text{ is a } \underline{\text{month}} \text{ of a } \underline{\text{year}} \text{ } \underline{\text{not having 31 days}}\}$

(vi) $F = \{x : x \text{ is a constant in the English alphabet which precedes } k\}.$
 $= \{B, C, D, E, G, H, J\}$

EXAMPLE



Which of the following sets are Finite or Infinite

- (a) The set of months in a Year. $\mathbb{F}$
- (b) $\{1, 2, 3, \dots\}$ $(\mathbb{IF})$
- (c) The set of positive integers greater than 100. $(\mathbb{IF})$
- (d) The set of prime numbers less than 99. $(\mathbb{F})$.

QUESTION



(PYQJKAUSE)

Cardinal number of an empty set is:

1 0 ✓✓

2 1 ✗

3 2 ✗

4 Infinite ✗

↓
 ϕ (A set with no elements)

$$\{x : x \in \mathbb{N} \text{ and } x < 0\} = \phi$$

EXAMPLE



Are the following pair of sets equal? Give reasons.

(a) $A = \{2, 3\}$, $B = \{x : x \text{ is a solution of } x^2 + 5x + 6 = 0\} = \{-2, -3\}$.

(b) $A = \{x : x \text{ is a Letter in the word FOLLOW}\} = \{F, O, L, W\}$
 $B = \{y : y \text{ is a Letter in the word WOLF}\}$

$$= \{W, O, L, F\}$$

$$\begin{aligned} & \rightarrow x^2 + 5x + 6 = 0 \\ & (x+2)(x+3) = 0 \\ & x = -2, -3 \end{aligned}$$

$$A \neq B$$

EXAMPLE



Which of the following are examples of the null set

- (i) Set of odd natural numbers divisible by 2
- (ii) Set of even prime numbers
- (iii) $\{x : x \text{ is a natural numbers, } x < 5 \text{ and } x > 7\}$
- (iv) $\{y : y \text{ is a point common to any two parallel lines}\}$

SUBSETS

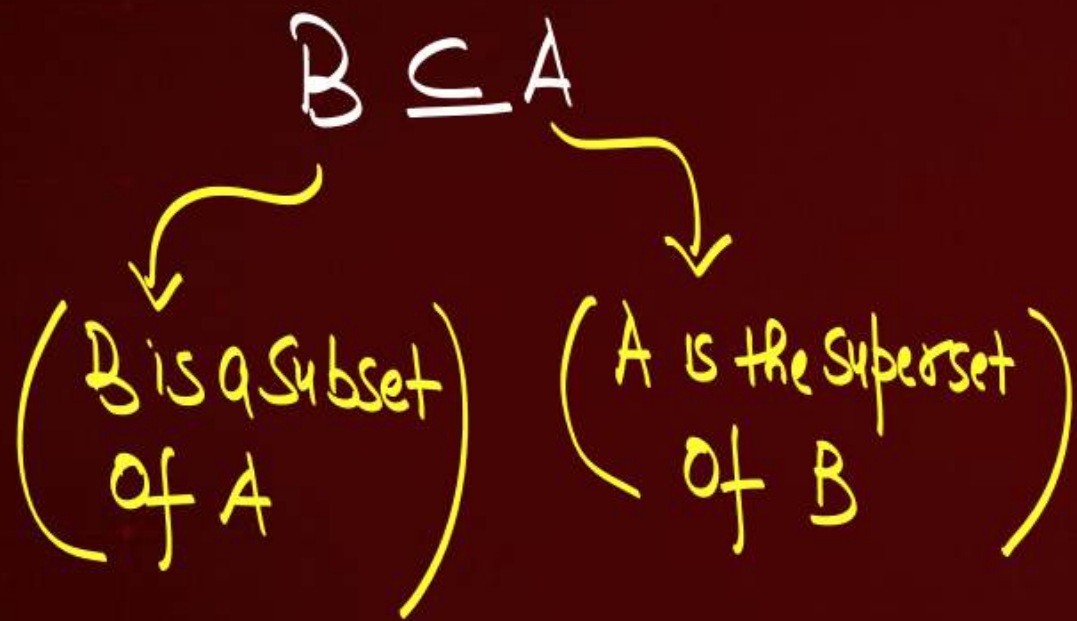


Definition: A set 'B' is said to be a Subset of a Set A if every element of B is Contained in A.

ϕ is the Subset of every A.

B is a Subset of A & $B \neq A$.

$\Rightarrow$ B is a Proper Subset of A.



$$A = \{1, 2, 3, 4, 5\} \longrightarrow 2^5 = 32.$$

$$B = \{1, 2, 3\}$$

B is a Subset of A.

- * NO. of Subsets of a Set A = $2^{n(A)}$
- * The Set of all the subsets of the Set A is known as the Power Set of A. ($P(A)$).

QUESTION



The set $\{1, 2, 3\}$ is the proper subset of

- 1 $\{1, 2\}$ ✓
- 2 $\{1, 3, 4\}$ ✗
- 3 $\{1, 2, 3\}$ ✗
- 4 $\{\{1, 2\}, 3, 4\}$ ✗

QUESTION



The set $A = \{1, 2, 3\}$ is the subset of

- 1 $\{1, 2, 4, 5\}$ ✗
- 2 $\{1, \{2, 3\}, 4, 5\}$ ✗
- 3 $\{1, 2, 3, 7, 8\}$ ✓✓
- 4 $\{\{1, 2\}, 3, 4\}$ ✗

QUESTION



If set $A = \{1, 2, 3\}$, then $P(A)$ is equal to

- 1 ☒ $\{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}\}$
- 2 ☒ $\{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{3, 1\}\}$
- 3 ☒ $\{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$
- 4 ☒ $\{\{1\}, \{2\}, \{3\}, \{1, 2, 3\}\}$

$$\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \\ A = \{1, 2, 3\}$$

$$P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \\ \{2, 3\}, A\}$$

QUESTION



Which of the following is True?

1 $\times a \in \{\{a\}, b\}$ $\{a\} \in \{\{a\}, b\}$

2 $\times \{b, c\} \subset \{a, \{b, c\}\}$
 $b \notin \{a, \{b, c\}\}$

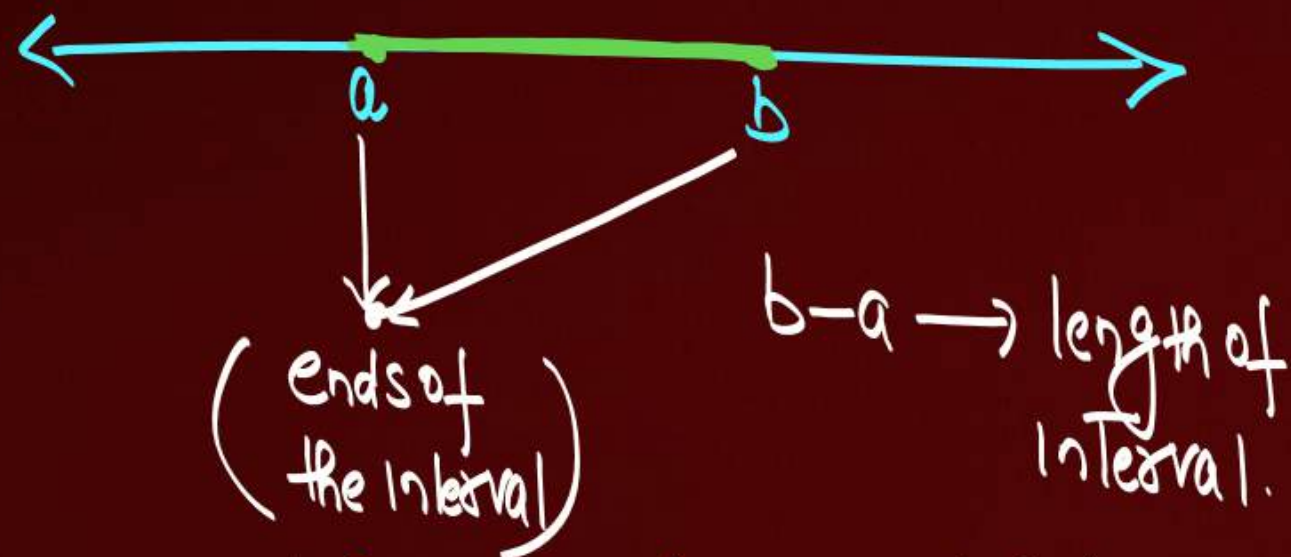
3 $\{a, b\} \subset \{a, \{b, c\}\}$

4 None of these

$$\{b, c\} \in \{a, \{b, c\}\}$$
$$\{\{b, c\}\} \subset \{a, \{b, c\}\}$$

Intervals

* Continuous part of a Real line is known as an Interval



* Both end points are included $[a, b]$ (Closed interval)

* Only one end is included $[a, b)$ / $(a, b]$ Semi-open / Semi closed interval

* Both are excluded (a, b) -> open interval

$$* (a, b) = \{x \in \mathbb{R} : a < x < b\}$$

$$* [a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

$$* (a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$

$$* [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

QUESTION



The subset of $\mathbb{R}$ as Intervals $\{x : x \in \mathbb{R}, -12 < x < -10\}$ is

1 $(-12, -10]$ ✗

2 $[-12, -10]$ ✗

3 $(-12, -10)$ ✓

4 $[-12, -10)$ ✗

QUESTION



Which of the following sets is the **Universal set** of sets

$A = \{2, 4, 5\}$, $B = \{1, 3, 5\}$, $C = \{3, 5, 7, 11\}$, $D = \{2, 4, 8, 10\}$

1 $\{1, 2, 3, 5, 6, 7, 8, 9, 10\}$

2 $\{1, 2, 3, 4, 5, 6, 7, 8, 10\}$

3 ✓✓ $\{1, 2, 3, 4, 5, 6, 7, 8, 10, 11\}$

4 $\{1, 2, 3, 4, 6, 7, 8, 10, 11\}$

QUESTION



Make correct statements by filling in the symbols $\subset$ or $\not\subset$ in the blank spaces:

- (i) $\{2, 3, 4\} \dots \{1, 2, 3, 4, 5\}$
- (ii) $\{a, b, c\} \dots \{b, c, d\}$ H/W
- (iii) $\{x : x \text{ is a student of Class XI of your school}\} \dots \{x : x \text{ student of your school}\}$
- (iv) $\{x : x \text{ is a circle in the plane}\} \dots \{x : x \text{ is a circle in the same plane with radius 1 unit}\}$
- (v) $\{x : x \text{ is a triangle in a plane}\} \dots \{x : x \text{ is a rectangle in the plane}\}$
- (vi) $\{x : x \text{ is an equilateral triangle in a plane}\} \dots \{x : x \text{ is a triangle in the same plane}\}$
- (vii) $\{x : x \text{ is an even natural number}\} \dots \{x : x \text{ is an integer}\}$

QUESTION



Write the following as intervals:

(i) $\{x : x \in \mathbb{R}, -4 < x \leq 6\}$ *HP/W*

(ii) $\{x : x \in \mathbb{R}, -12 < x < -10\}$

(iii) $\{x : x \in \mathbb{R}, 0 \leq x < 7\}$

(iv) $\{x : x \in \mathbb{R}, 3 \leq x \leq 4\}$

QUESTION



Write the following intervals in set-builder form:

(i) $(-3, 0)$

h/w

(ii) $[6, 12]$

(iii) $(6, 12]$

(iv) $[-23, 5)$

QUESTION



What universal set(s) would you propose for each of the following:

- (i) The set of right triangles.
- (ii) The set of isosceles triangles.

h/w

QUESTION



Given the sets $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{0, 2, 4, 6, 8\}$, which of the following may be considered as universal set (s) for all three sets A, B and C

(i) $\{0, 1, 2, 3, 4, 5, 6\}$

(ii) ϕ *h/w*

(iii) $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

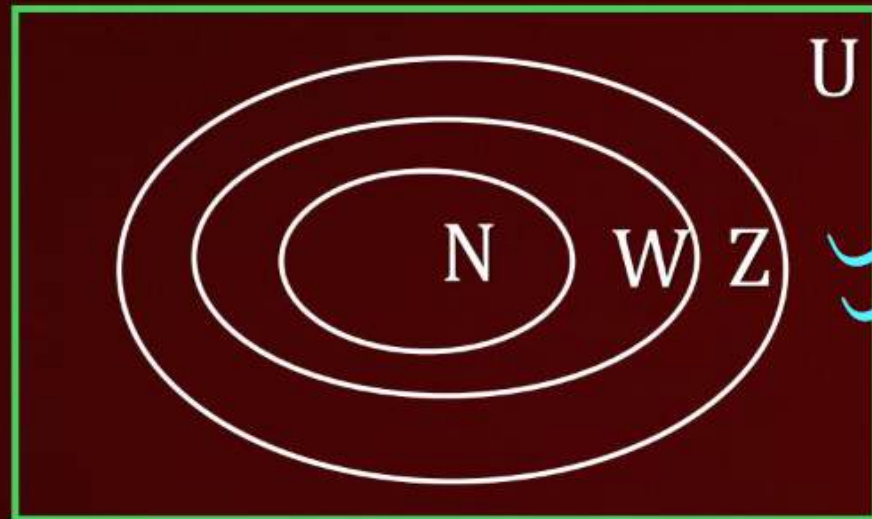
(iv) $\{1, 2, 3, 4, 5, 6, 7, 8\}$

QUESTION



The set of natural numbers is a subset of whole numbers which is a Subset of Integers.
Then, correct representation of it through Venn Diagram is:

1



2



3



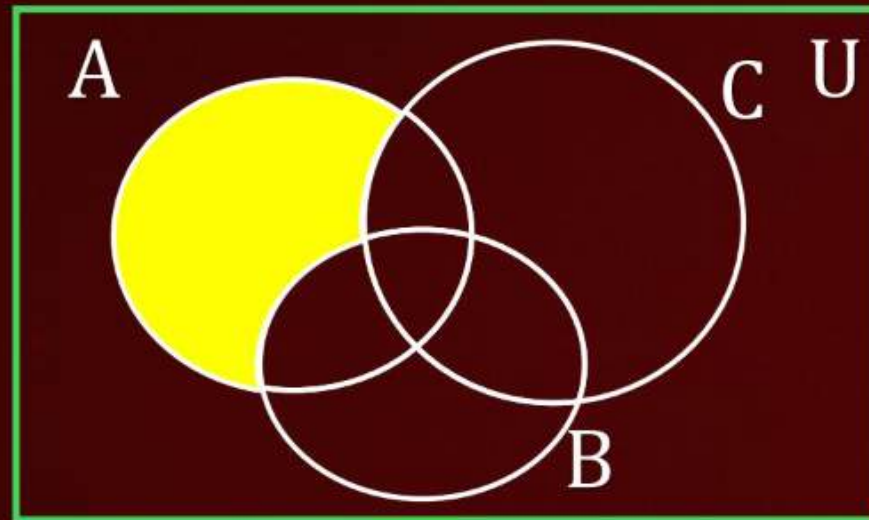
4

None of These

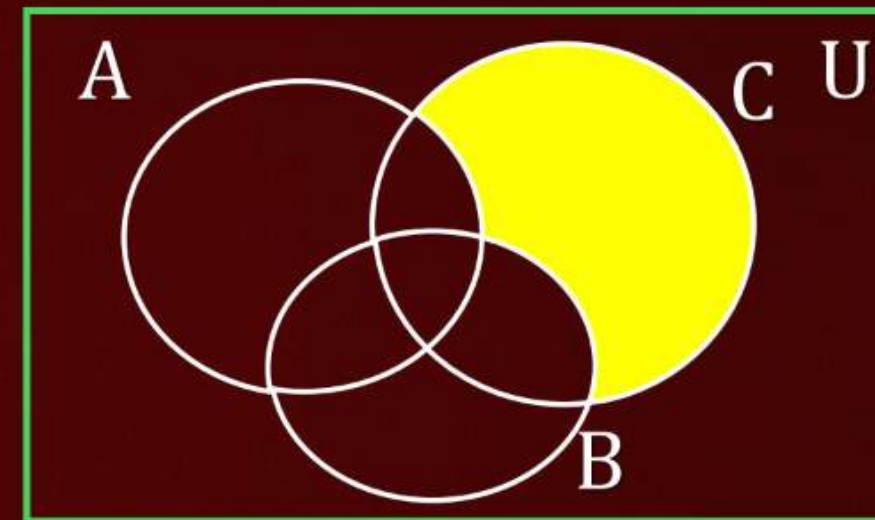
QUESTION

Out of the following Venn Diagrams, which represents only A, given three sets A, B and C?

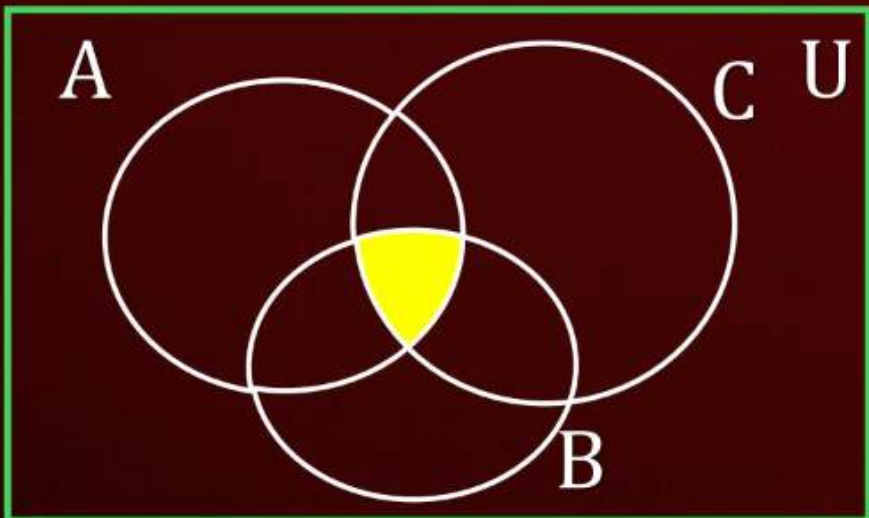
~~1~~



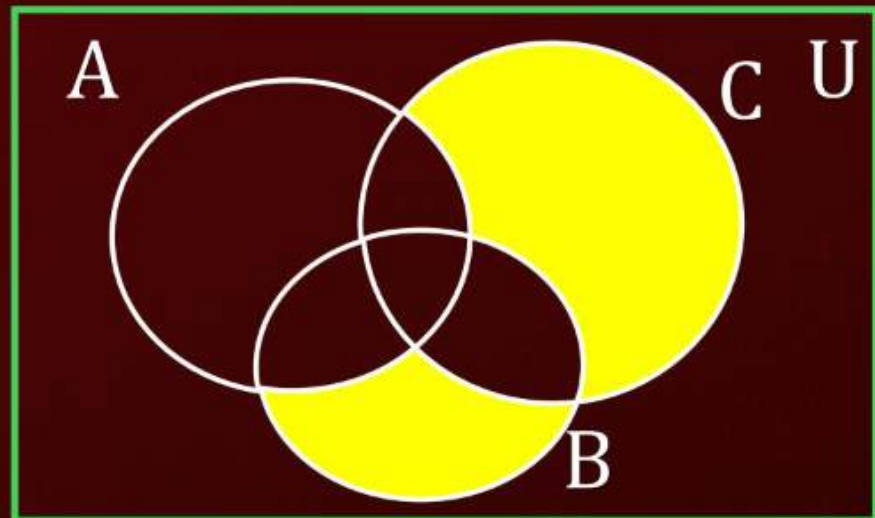
~~2~~



~~3~~



~~4~~



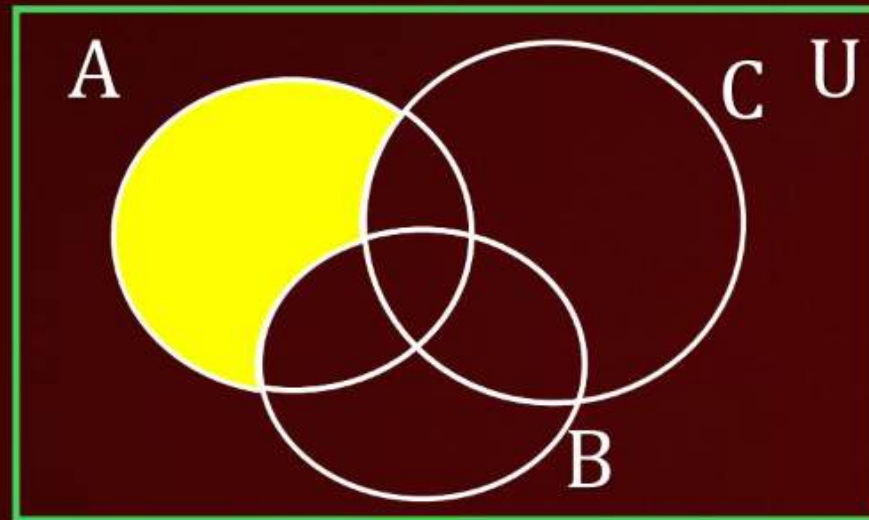
QUESTION



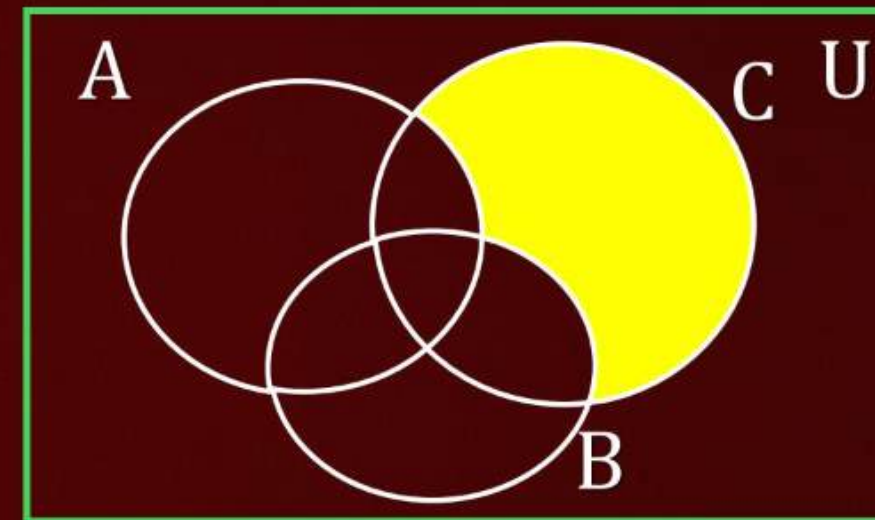
Out of the following Venn Diagrams, which represents the region common to all the sets A, B and C?

H/W

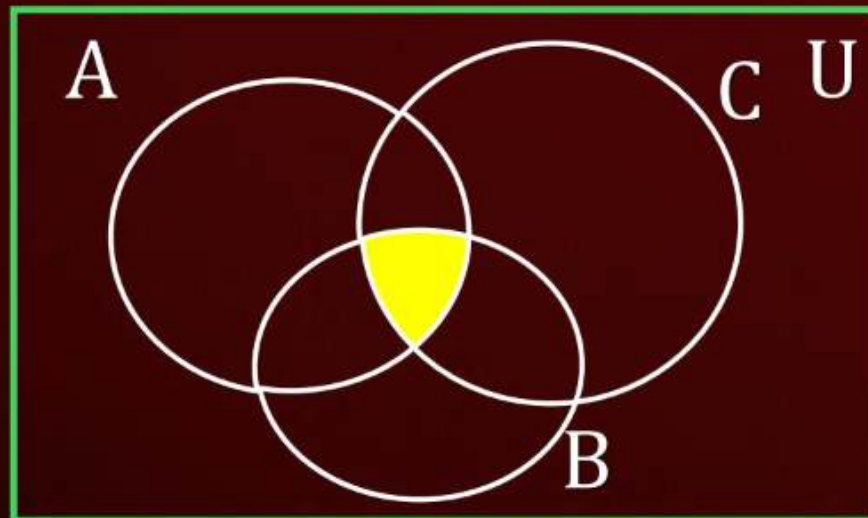
1



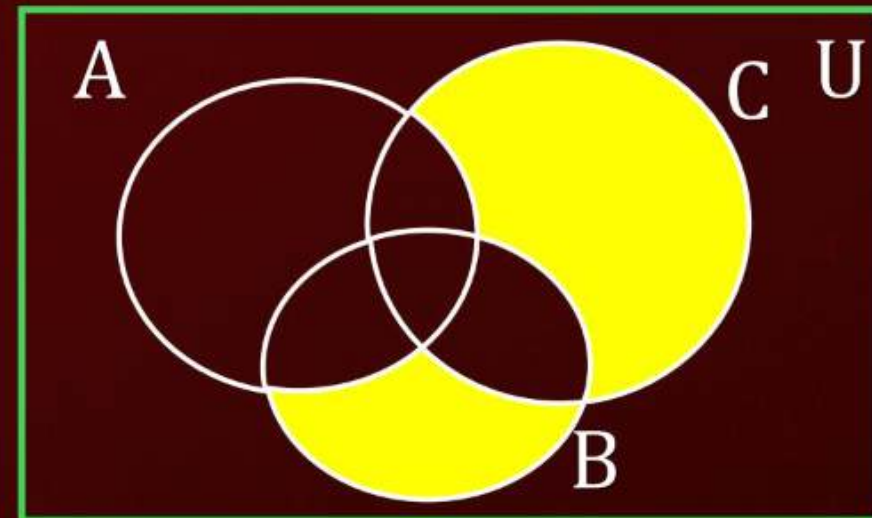
2



3



4



QUESTION



If two sets $A = \{1, 2, 3\}$ and $B = \{1, 3, 5, 7\}$ then $A \cup B$ is equal to

$$\{1, 2, 3, 5, 7\}$$

- 1 $\{1, 2, 3, 7\}$ ✗
- 2 $\{1, 2, 3, 5, 7, 8\}$ ✗
- 3 $\{1, 2, 3, 5, 7\}$ ✓
- 4 $\{1, 2, 3, 4, 5, 6, 7\}$ ✗

QUESTION



If two sets $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 5\}$, then $A \cap B$ is equal to

1 $\{1, 3\}$ ✗

2 $\{2, 4\}$ ✓

3 $\{4, 3\}$ ✗

4 $\{1, 6\}$ ✗

$$\{2, 4\}$$

QUESTION



Given that $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{7, 8, 9, 10, 11\}$, $C = \{6, 8, 10, 12, 14\}$. Which of the following pair of sets are Disjoint sets?

→ ϕ (Intersection)

$$A \cap B = \phi$$

- 1 A and B ✓✓
- 2 B and C ✗
- 3 C and A ✗
- 4 None of the above

QUESTION



If $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $B = \{1, 4, 7, 8, 9\}$ then $A - B$ equals to

- 1 $\{1, 2, 3, 4, 5\}$ $A - B = \{2, 3, 5, 6\}$
- 2 $\{2, 3, 6, 4, 5\}$
- 3 $\{2, 3, 5, 6\}$
- 4 None of the above

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

QUESTION



If R is the set of Real numbers and Q is the set of Rational numbers, then $R-Q$ is

- 1 The set of Integers ✗
- 2 The set of Non negative Real numbers ✗
- 3 The set of Irrational Numbers ✓
- 4 The set of negative Real numbers ✗

QUESTION



If $A = \{3, 5, 7, 9, 11\}$, $B = \{7, 9, 11, 13\}$, $C = \{11, 13, 15\}$ and $D = \{15, 17\}$; find

- (i) $A \cap B$
- (ii) $B \cap C$
- (iii) $A \cap C \cap D$
- (iv) $A \cap C$
- (v) $B \cap D$
- (vi) $A \cap (B \cup C)$
- (vii) $A \cap D$
- (viii) $A \cap (B \cup D)$
- (ix) $(A \cap B) \cap (B \cup C)$
- (x) $(A \cap D) \cap (B \cup C)$

QUESTION



If

$A = \{x : x \text{ is a natural number}\},$

$B = \{x : x \text{ is an even natural number}\}$

$C = \{x : x \text{ is an odd natural number}\}$

$D = \{x : x \text{ is a prime number}\},$ find

(i) $A \cap B$

(ii) $A \cap C$

(iii) $A \cap D$

(iv) $B \cap C$

(v) $B \cap D$

(vi) $C \cap D$

QUESTION

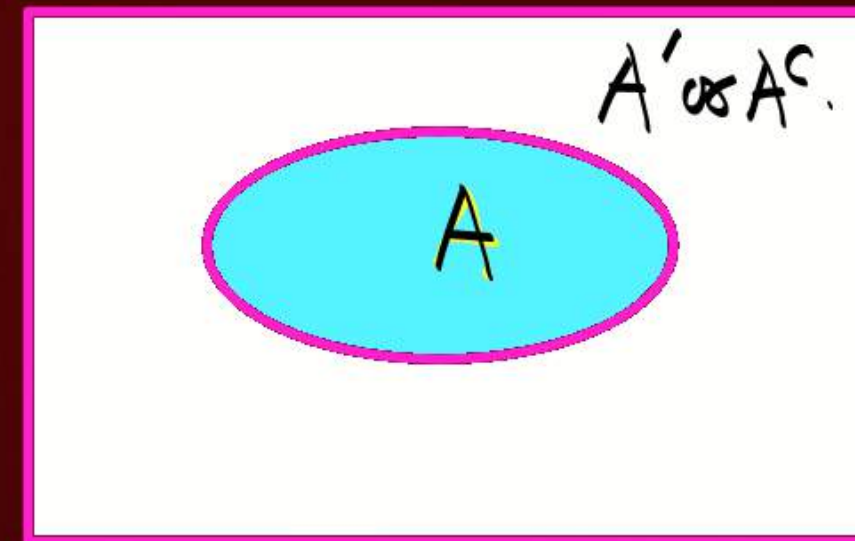


State whether each of the following statement is true or false. Justify your answer.

- (i) $\{2, 3, 4, 5\}$ and $\{3, 6\}$ are disjoint sets.
- (ii) $\{a, e, i, e, i, o, u\}$ and $\{a, b, c, d\}$ are disjoint sets.
- (iii) $\{2, 6, 10, 14\}$ and $\{3, 7, 11, 15\}$ are disjoint sets.
- (iv) $\{2, 6, 10\}$ and $\{3, 7, 11\}$ are disjoint sets.

Complement of a Set

$$A^c \text{ or } A' = \{x : x \in U \text{ but } x \notin A\} = U - A.$$



Properties of Complement Sets

(i) Complement Laws:

$$(a) \quad A \cup A' = U$$

$$(b) \quad A \cap A' = \phi$$

(ii) De Morgan's laws:

$$(b) \quad (A \cup B)' = A' \cap B'$$

$$(b) \quad (A \cap B)' = A' \cup B'$$

(iii) Law of Double Decomposition:

$$(A')' = A$$

(iv) Law of empty set and universal set:

$$\phi' = U \text{ and } U' = \phi.$$

QUESTION



If $U = \{a, b, c, d, e, f, g, h\}$, find the complements of the following sets:

(i) $A = \{a, b, c\} \longrightarrow A' = \{d, e, f, g, h\}$

(ii) $B = \{d, e, f, g\} \quad B' = \{a, b, c, h\}$

(iii) $C = \{a, c, e, g\}$

(iv) $D = \{f, g, h, a\}$

QUESTION

(PYQ JKBOSE)

If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$ and $B = \{2, 3, 5, 7\}$ Verify that

- (i) $(A \cup B)' = A' \cap B'$ (De-Morgan laws)
(ii) $(A \cap B)' = A' \cup B'$

(Sol) (i) Here $A \cup B = \{2, 3, 4, 5, 6, 7, 8\}$

$$\Rightarrow (A \cup B)' = \{1, 9\} \text{--- (1)}$$

$$\text{Also } A' = \{1, 3, 5, 7, 9\}$$

$$\& B' = \{1, 4, 6, 8, 9\}$$

$$\Rightarrow A' \cap B' = \{1, 9\} \text{--- (2)}$$

From (1) & (2), we get;

$$(A \cup B)' = A' \cap B'. \text{ Hence the result.}$$

QUESTION



Let U be the set of all triangles in a plane. If A is the set of all triangles with at least one angle different from 60° , what is A' ?

H/w

QUESTION



Fill in the blanks to make each of the following a true statement:

(i) $A \cup A' = \dots$

(ii) $\phi' \cap A = \dots = A$

(iii) $A \cap A' = \dots$

(iv) $U' \cap A = \dots = \phi$



QUESTION

(PYQ TK BOSE)

Let A, B, and C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$.

Show that $B = C$.

Proof: We know that

$$B = B \cap (A \cup B)$$

$$\Rightarrow B = B \cap (A \cup C) \quad (\text{Given } A \cup B = A \cup C)$$

$$\Rightarrow B = (B \cap A) \cup (B \cap C) \quad (\text{Distributive law})$$

$$\Rightarrow B = (A \cap B) \cup (B \cap C)$$

$$\Rightarrow B = (A \cap C) \cup (B \cap C) \quad (\text{Given } A \cap C = A \cap B)$$

$$\Rightarrow B = (C \cap A) \cup (C \cap B)$$

$$\Rightarrow B = C \cap (A \cup B) \quad (\text{Distributive law})$$

$$\Rightarrow B = C \cap (A \cup C) \quad (\text{Given } A \cup B = A \cup C)$$

$$\Rightarrow B = C$$

Hence proved.

QUESTION



Show that $A \cap B = A \cap C$ need not imply $B = C$.

h/w.

QUESTION



Let A and B be sets. If $A \cap X = B \cap X = \phi$ and $A \cup X = B \cup X$ for some set X , show that $A = B$.

(**Hints:** $A = A \cap (A \cup X)$, $B = B \cap (B \cup X)$ and use Distributive law)

H/W

QUESTION



Find sets A , B and C such that $A \cap B$, $B \cap C$ and $A \cap C$ are non-empty sets and $A \cap B \cap C = \phi$.

h/w

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Thank
You